Joshua Lollis

Fullerton, CA | (608) 397-7678 | lollisjosh@csu.fullerton.edu
Portfolio | GitHub | LinkedIn

Career-changing full-stack SWE with 10+ years developing personal and academic projects spanning dev tools, music production utilities, desktop apps, and websites, with a special focus on games. Recently graduated with a B.S. in Computer Science from CSUF (Cum Laude, GPA: 3.63) after a non-traditional educational journey.

Education

B.S. Computer Science (GPA: 3.63, Cum Laude) Aug 2020−Dec 2025
California State University, Fullerton *Degree conferred Jan 2, 2026 [Click here for e-diploma]*
A.S. Computer Science (Honors) Jan 2015−May 2020
A.A. Interdisciplinary Math & Science (Honors) Jan 2015−May 2020
Fullerton College

Coursework

OOP (C++) • Computer Graphics • C# Programming • Game Design • Senior Capstone • Data Struct. • Software Eng. Principles • Algo. Eng. • OS • AI • File Struct. & Databases • Open Source Dev. • Android Dev. • Web Front-End Eng. • Computer Org. & Assembly • Compilers & Lang. • Lin. Algebra & Graph Theory

Skills

[Click here for recent skills telemetry]

Stack: C/C++ • Python • C# • Unity Engine • HTML • CSS • JavaScript
Env: Linux • Windows • Android • VS Code
Tools: Git • CMake • GitHub (Issues, PRs, Projects)

Projects

[Click project names for more info]

Hardware Interface Utility: **XB2MIDI** *(Personal, Solo)* [C#, .NET, WPF/MVVM, NAudio, XInput]
Windows tool that maps Xbox controller inputs to MIDI messages/macros for DAW/VST workflows.

- Built a low-latency control input pipeline with debouncing and thread-safe cursor to MIDI translation.
- Created an extensible JSON-backed mapping system for modes/profiles and custom bindings.
- Implemented MIDI device lifecycle handling (enumeration, hot-swap, reconnect, rate limiting).

Full-Stack Game Software: **Antillery** *(Capstone, Solo)* [Unity, C#, UXML, USS, Addressables]
2D artillery game with async scene loading and ScriptableObject-driven state machines.

- Built an async loading framework using Addressables + ScriptableObject-based states.
- Implemented the match loop (turn alternation, timers, damage, win/loss).
- Structured unit behavior via composition and a factory-built state machine.

Embedded Firmware: **Pico Telemetry Node** *(Personal, Solo)* [C/C++, Unity Test, Pico, CMake, Python]
Pico telemetry/command node with a framed serial protocol, streaming parser, and state machine.

- Implemented framed serial packets with streaming parse, CRC16 validation, and auto-resync on malformed frames.
- Built BOOT/IDLE/ARMED/FAULT state machine with fault handling and HAL abstraction for testability.
- Wrote Python host tool for command injection and telemetry logging; unit-tested state logic using Unity/CMake.

Full-Stack Game Software: **Snake** *(Personal, Solo)* [C++, SFML, CMake, State/Command/Observer]
Complete desktop game with menus, gameplay states, pause, persistence, and new mechanics.

- Implemented menu + gameplay state system and decoupled input routing using State/Command patterns.
- Architected gameplay into discrete systems (movement, collision, scoring) to improve maintainability and testing.
- Persisted high scores using JSON serialization across sessions.

API-Driven Desktop App: **Card Collection** *(Academic, Team Lead)* [Python, Qt/QML, JS, REST, Agile]
Desktop app for searching and collecting Pokémon TCG cards via a public API.

- Led a small team through sprint/roadmap and delivered working milestones.
- Implemented async API search over 50k+ cards using Python signals/slots for a responsive UI.
- Built a modular UI with 20+ QML components, using JavaScript for UI logic and Python signals/slots for app wiring.

Responsive Static Site: **Portfolio Site** *(Academic → Personal, Solo)* [HTML, CSS, JavaScript, Chart.js]
Personal portfolio site showcasing projects, skills, and experience with a responsive, interactive UI.

- Actively maintained and iterated on since 2024, adding features and refining UX as new projects are shipped.
- Built reusable UI via component injection; added smooth scrolling, micro-interactions, and an image modal.
- Implemented mobile-first layouts with separate mobile/desktop CSS and device-specific interactions (hover vs. touch).

Experience

2007−Present

Please see LinkedIn for full employment history, spanning QA/QC, teaching, photography, graphic design, sales, and logistics.